Automatic Extraction of Electoral Results from Vote-Counting Records of the 2026 Peruvian General Elections

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***Abstract*—In Peru, official electoral results are published through vote-counting records, many of which are scanned. This hinders the automatic processing of information at scale. We present a complete system that downloads records from the National Office of Electoral Processes (ONPE) portal, locates 46 fields in each document using a template and printed reference marks, and recognizes handwritten counts with optical character recognition (OCR) restricted to digits. The evaluation did not require manual labeling: the officially digitized and published ONPE results were used as the reference. The test set consisted of a random sample of 100 records from 23 departments. Twenty-four percent of the records were digital documents with an extractable text layer, achieving 95.1% field-level accuracy. For the 76 scanned handwritten records (3,268 fields), the system achieved 59.6% field-level accuracy and a Character Error Rate (CER) of 0.401. On the same records, a multimodal Large Language Model (LLM) achieved 48.4%. The results show that a specialized, multi-stage system is a useful foundation for improving handwritten-digit recognition in electoral documents.**

***Index Terms*—document analysis, optical character recognition, handwritten digit recognition, electoral documents, template alignment**

I. Introduction

Vote-counting records contain the official results for every polling station in Peru. Although ONPE publishes these documents digitally, much of the information is contained in scanned forms in which polling-station officials wrote the votes by hand. Reviewing these records manually is slow, can introduce errors, and is not feasible for the approximately 88,000 polling stations in a general election. This limits automated auditing and independent verification of the results.

This work addresses the following question: *how accurate can a traditional computer-vision system be when extracting results from Peruvian records selected at a national scale, and what does this imply for a real-world system?* To answer it, we built and evaluated a five-stage process: downloading, image preparation, field localization, digit recognition, and generation of structured output. The study used records from the first round of the 2026 General Elections. The main contributions are as follows:

1) **A reproducible evaluation without manual labeling.** The ONPE portal publishes both the image of each record and the officially digitized counts. These values were used as the reference for evaluating the system. Thus, each record provides approximately 43 already-verified field–value pairs. The test sample was randomly selected nationwide using a published seed.
2) **A template aligned using printed form marks.** The record has a fixed layout and contains black squares along its edges. These marks make it possible to correct displacements and small distortions introduced during scanning. This correction improved accuracy by 8.7 percentage points compared with using fixed coordinates without alignment.
3) **Identification of two record types.** Twenty-four percent of the sample consists of digitally generated records whose text layer can be directly extracted and achieves 95.1% field-level accuracy. The remaining 76% are scanned handwritten records that require image processing. Therefore, a real-world system must first identify the document type and apply the appropriate process.
4) **A clear measure of general-purpose OCR performance.** On handwritten records, digit-restricted OCR achieved 59.6% field-level accuracy. This result shows the limitation of using a general recognizer without training it on this document type and supports the need to train a specialized model.
5) **A comparison with a multimodal language model.** Under the same records and evaluation conditions, a multimodal model given the complete page and two solved examples achieved 48.4%, compared with 60.9% for the multi-stage system. The multimodal model offers a useful initial approach, but it does not outperform the specialized method for reading handwritten digits.

The code, field-level results, and sampling seed are publicly available.[1]

II. Related Work

Object detection. RT-DETR [1] is a real-time object-detection model that, according to its authors, outperforms similarly sized YOLO models in accuracy and speed. In vote-counting records, where cells are close together and may be covered by stamps or signatures, learned detectors could be an alternative for locating fields. In this work, field localization is solved with a template aligned to the document. In the future,

[1] https://github.com/carlosperez100/onpe_actas

this template could also automatically generate labels to train detectors such as YOLOv11 or RT-DETR.

Text recognition. TrOCR [4] uses *transformer* neural networks for optical character recognition. DTrOCR [2] proposes a variant for printed, scene, and handwritten text recognition. These studies support the use of specialized models to improve handwritten-digit reading. In our case, it is important to prevent a model from changing a number into a seemingly more likely sequence, because each value represents an electoral count.

End-to-end document analysis. OmniParser [3] integrates text localization, key information extraction, and table recognition. Donut [5], by contrast, seeks to generate the document structure directly from the image. Results from these works show that documents with extensive tables can be difficult to process in a single step. We therefore retain a multi-stage process, as it enables clear inspection and measurement of the performance of each system component. End-to-end methods are considered an alternative for future work.

The reviewed studies do not specifically evaluate handwritten digits in Peruvian electoral records containing stamps and signatures. This is the practical need addressed by this study.

III. Dataset

A. Source and download

All documents come from the official ONPE results portal for the first round of the 2026 General Elections, held on April 12, 2026. The portal allows users to query the records for a polling station, their processing status, and associated files. Each vote-counting record is downloaded as a PDF from a temporary address provided by the portal. Downloads were performed at a moderate rate and logged in a file that allows them to be resumed after an interruption (`src/scraper/`).

B. Sample design

The study population contains approximately 88,064 polling-station codes, estimated through queries to the portal services. A simple random sample without replacement of 100 records was drawn using a published seed (2026). The sample covers 23 departments, identified using *ubigeo* prefixes, where *ubigeo* is the standard geographic code used in Peru. Lima accounted for 36%, consistent with its greater weight in the electorate. This design avoids the bias of selecting polling stations with consecutive codes from a single district.

C. Record types

Reviewing the number of pages and the PDF text layer identified two document types: **76 handwritten records**, which are one-page scans with no extractable text, and **24 digital STAE records**, which have two pages, a digital signature, and counts written in a digital typeface. STAE stands for *Solución Tecnológica de Apoyo al Escrutinio* (Technological Solution for Supporting Vote Counting), an ONPE tool that helps polling-station officials record and print the records. In this study, STAE records are the documents digitally generated using that tool.

The proportion of digital records in the sample is 24%, with a 95% Wilson confidence interval from 16.7% to 33.2%. This result confirms that the population includes both document types, although the sample is not intended to estimate their exact proportion. Vision experiments were performed using the 76 handwritten records; digital records were processed through direct extraction of the text layer (Sec. IV-E).

D. Reference values

For each polling station in the sample, the counts officially digitized by ONPE were saved: votes by political organization, blank, null, and challenged votes, totals, and the number of voters. Positions 04 and 14 of the form correspond to withdrawn parties; they are therefore empty in the record and have no value in the official data. They were excluded from the evaluation to avoid counting matches between empty fields as correct predictions. In total, each record provides 43 evaluable fields, for a total of 3,268 fields in handwritten records.

IV. Method

A. Overview

The system comprises five stages: (1) downloading the PDF and official reference values; (2) image preparation; (3) region-of-interest localization; (4) value recognition; and (5) generation of a structured file for each record. Image preparation includes converting the PDF to an image at 300 pixels per inch, correcting skew, improving contrast, and reducing scan noise.

B. Field localization using a template and reference marks

The record has a fixed format. Therefore, 46 regions were defined once in a reference record: 38 vote cells, four summary rows, and four header fields. The coordinates were stored as proportions of page width and height (Fig. 1).

The direct use of fixed coordinates fails when the image contains black margins, displacements, or slight distortions introduced by the scanner. In some records, the content was shifted, causing a cell to be cropped over the wrong row. To solve this, the black squares printed along the form edges are used as reference marks. First, dark squares near the borders are detected. They are then compared with 15 known positions in the reference record to compute the most consistent displacement, ignoring missing marks or false detections. Finally, a simple geometric correction is applied to the page before cropping the fields.

C. Recognition of handwritten counts

Each cell is processed with EasyOCR, configured to recognize digits only. Before recognition, improvements obtained from an error analysis of 10 pilot records are applied (Fig. 2). First, an inner margin is removed to prevent the cell's printed borders from being interpreted as digits. The image is then converted to black and white using a threshold computed for each cell, so that handwritten ink is highlighted. Dotted separators printed at known cell positions and small scanning artifacts are also removed. In fields containing one box per

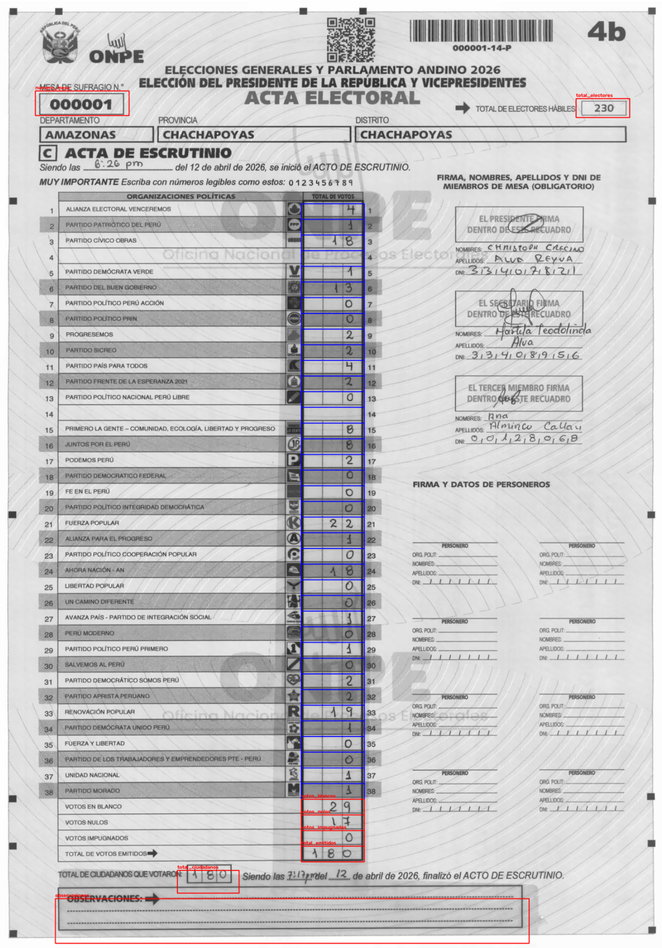

Figure 1. Reference record with the 46 template regions: 38 vote cells (blue), summary and header fields (red). Coordinates are applied after aligning the document using the reference marks.

digit, each box is read separately. If this process removes overly faint strokes, the system retries recognition using the original crop.

D. Rule for empty cells

Polling-station officials often leave blank the cell of a party that received no votes. In this form, an empty cell represents zero and does not necessarily indicate an OCR error. Therefore, when no value is recognized in a party cell or in challenged votes, the system assigns zero. This contribution of form-specific knowledge is reported separately to distinguish it from the image-recognition performance itself.

E. Processing of digital records

For digital STAE records, the results are extracted directly from the PDF text layer. Each party name appears next to its count, and summary labels appear before their values. Extracted names are compared with the official names after normalizing capitalization, spaces, and accents. Thus, no image processing is required.

Table I
CUMULATIVE EFFECT OF IMPROVEMENTS IN THE 10-RECORD PILOT.

Configuration	Accuracy (%)	CER
v1: digit OCR without cleaning	43.95	0.524
v2: + ink, border, and box cleaning	52.09	0.419
v3: + separator and artifact removal	52.79	0.476
v4: + empty-cell = zero rule	58.37	0.420

Table II
RESULTS ON THE NATIONWIDE RANDOM SAMPLE OF 76 HANDWRITTEN RECORDS.

Configuration	Accuracy (%)	CER
Fixed template, OCR without cleaning	26.81	0.724
Fixed template + zero rule	50.86	0.482
Aligned template, OCR without cleaning	38.00	0.617
Aligned template + zero rule	**59.58**	**0.401**

V. EXPERIMENTS

A. Metrics and statistical procedure

Field-level accuracy is the percentage of fields whose read value exactly matches the official value. **CER** (*Character Error Rate*) is the character-level error rate: it compares recognized digit strings with the correct strings. A lower CER indicates fewer errors.

Fields from the same record may be related because they share the same scan quality and handwriting. For this reason, confidence intervals were calculated by resampling complete records, with 10,000 repetitions. Field-level results and per-record metrics are available in the project repository.

B. Incremental evaluation of improvements in the pilot (10 records)

In an initial analysis, consecutive polling stations from a single district were used. Therefore, it does not represent the country, but it does allow us to measure the contribution of each improvement added to recognition (Table I).

C. National evaluation (76 handwritten records)

Table II presents the system results on the nationwide random sample. Alignment through reference marks proved essential. Without it, framing variations severely affected some records: the worst one achieved 14% accuracy and a CER above 1. With alignment, accuracy increased by 8.7 percentage points, and the worst record rose from 14% to 30%. The empty-cell-equals-zero rule added 21.6 percentage points, as these cases were frequent in the national sample.

Main result. The system achieved **59.58% field-level accuracy**, with a 95% confidence interval from 56.61% to 62.48%. CER was **0.401**, with a 95% confidence interval from 0.360 to 0.444. Mean per-record accuracy was 59.6%, with a standard deviation of 13.0 percentage points and a range from 30% to 84%. The remaining differences are mainly due to handwriting and scan quality rather than field localization.

Digital records. Text-layer extraction from the 24 STAE records achieved **95.12%** accuracy (955 of 1,004 fields).

Table III
COMPARISON BETWEEN THE MULTI-STAGE SYSTEM AND A MULTIMODAL MODEL WITH TWO EXAMPLES, ON THE SAME 15 HANDWRITTEN RECORDS.

Method	Accuracy (%)	CER
Multi-stage system (aligned template + OCR)	**60.93**	**0.395**
Multimodal LLM with two examples	48.37	0.591

Remaining errors are related to matching party names, not to reading numbers. The 35.5 percentage point difference from handwritten records confirms the importance of identifying the document type before processing it.

D. Comparison with a multimodal language model

As an alternative to the multi-stage process, we evaluated a large language model with the ability to analyze images and text (LLM). The model received the complete image of each record and two example records with their structured outputs. It had to return the counts without using a template, alignment, or independent OCR. Both methods were compared on the same 15 handwritten records. The model used was Gemini 2.5 Flash, accessed on its free tier in July 2026. The code is available in `src/recognition/llm_ocr.py`.

The multi-stage system outperformed the multimodal model by 12.6 percentage points. The model correctly identified the form's overall structure, but had greater difficulty reading handwritten digits in low-quality scans. On the two records with the worst results, it achieved 20.9% and 23.3%, whereas the multi-stage system achieved 34.9% and 53.5%, respectively. This is because the OCR receives an already isolated cell, whereas the multimodal model must interpret the entire page and its visual noise. The multimodal model may be useful for preparing an initial set of labels for a new format, but it does not replace a specialized recognizer for this task.

E. Error analysis

Errors were concentrated in four situations: thin digits that OCR does not detect; handwritten strokes that extend into areas where separators are removed; confusions between similar shapes, such as 8 and 18 or 7 and 9; and fields made up of several boxes. These are mainly recognition errors rather than field localization errors. Therefore, the next step is to train a recognizer specific to these digits and this form.

VI. CONCLUSIONS AND FUTURE WORK

The proposed system achieved 59.6% field-level accuracy on handwritten records selected nationwide and 95.1% on digital records. The evaluation was reproducible and did not require manual labeling, as it used the results officially digitized by ONPE as the reference.

The study yields four main conclusions. First, before cropping fields with a template, the document must be aligned with its printed reference marks. Second, the rule that an empty cell equals zero must be distinguished from the strictly visual performance of OCR. Third, it is advisable to identify whether the document is handwritten or digital before selecting the processing method. Fourth, a general multimodal model requires less initial setup, but achieved 12.6 percentage points less accuracy than the specialized system.

As future work, we propose training a *transformer*-based recognizer with the field–value pairs obtained from the records (approximately 43 per record and 43,000 from 1,000 records). We also propose training field detectors, such as YOLOv11 and RT-DETR [1], using labels automatically created with the aligned template; comparing the multi-stage system with end-to-end document analysis methods [3], [5]; and using the multimodal model to support the generation of initial labels that can subsequently be reviewed by a person.

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Figure 2. Cleaning of problematic cells (top: original crop; bottom: processed crop). Borders and dotted separators that could be mistaken for digits are removed.